

Uninterruptible 5 V Supply with 11.1 V Li-ion Backup

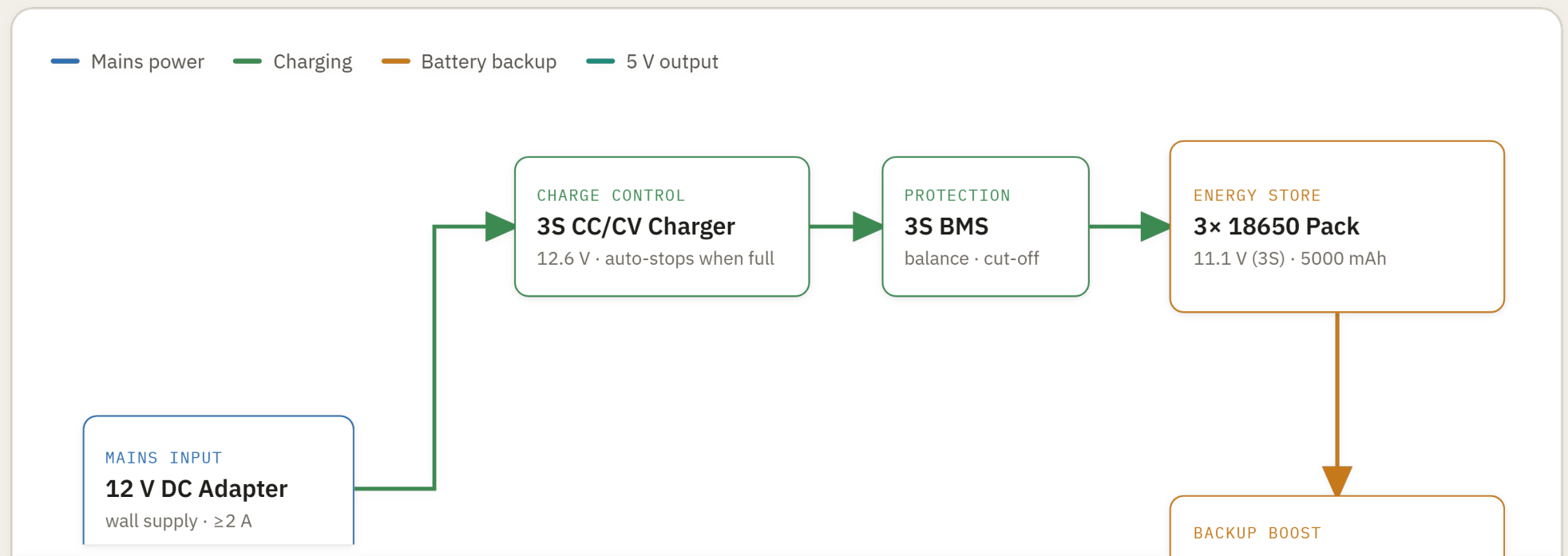
Runs your device from the wall while keeping the battery topped up, then switches to battery automatically the instant mains power is lost — with no interruption to the output.

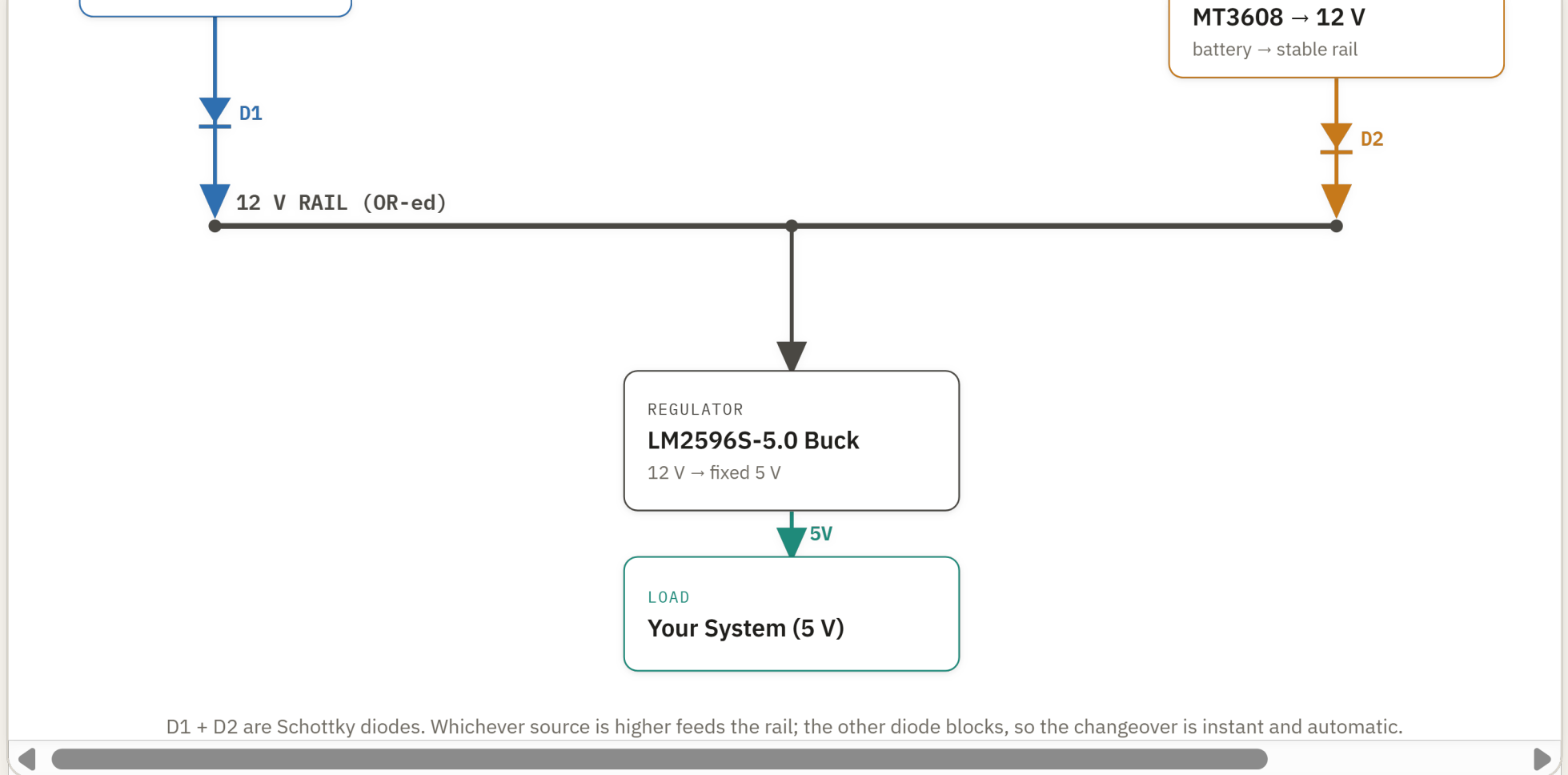


01 — BLOCK DIAGRAM

How the modules connect

Each labelled box is a ready-made module. Follow the arrows: blue is mains power, green is charging, amber is the battery backup path.





02 — WORKING PRINCIPLE

Three operating states

01 Mains present — normal run

The 12 V adapter does two jobs at once: it feeds the rail through **D1** to power your device via the buck, **and** it charges the battery through the charger + BMS. **D2 is blocked** because the rail (mains) sits higher

02 Battery full — charging stops

When the pack reaches **12.6 V**, the CC/CV charger tapers its current to zero and holds — it stops charging on its own. The BMS is the safety backstop, cutting in if any cell ever

03 Mains fails — battery backup

The rail voltage starts to fall, so **D2 now conducts**: the MT3608 boosts the battery up to a steady 12 V and feeds the rail through D2. The buck never notices the difference, so your 5 V output stays on with **no**

than the boost output, so the battery just sits and charges.

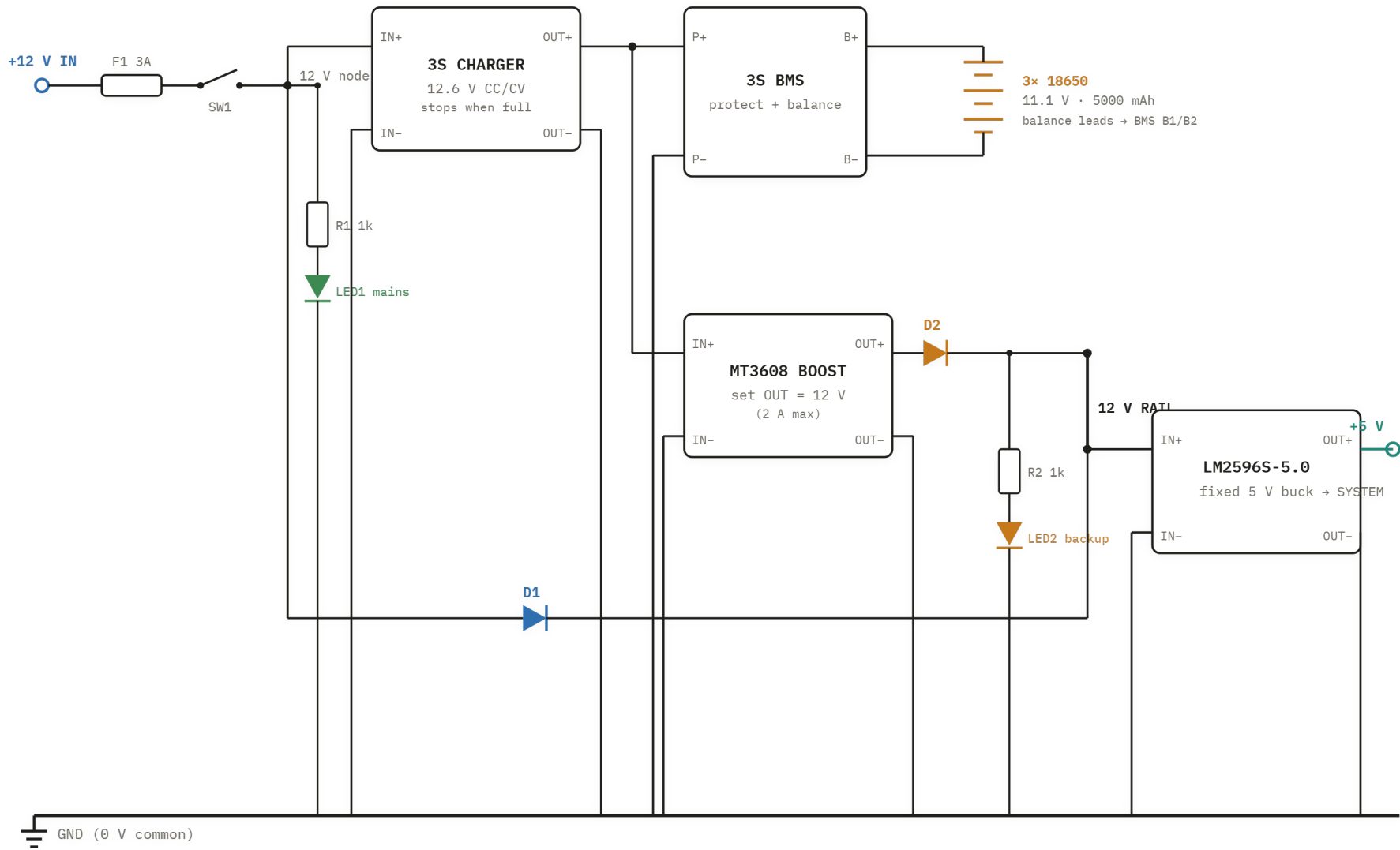
rises too high. Your device keeps running on mains the whole time.

interruption. When mains returns, D1 takes over again and charging resumes.

03 – SCHEMATIC

Wiring schematic (pin level)

Modules are shown as boxes with their real terminal names. Every black line is a wire; all the “–” terminals tie to the common ground bus along the bottom.



Where two wires cross without a solid dot, they are **not** connected. Dots mark real junctions.

04 — BILL OF MATERIALS

Parts list

REF	COMPONENT	SPEC / SETTING	QTY	PURPOSE
ADP	DC adapter	12 V, ≥2 A	1	Wall power (12.6–14 V for full charge)
CHG	3S Li-ion charge module (CC/CV)	12.6 V, ~2 A	1	Regulates charge & auto-stops when full
BMS	3S BMS / protection board	12.6 V, balanced, ≥8 A	1	Over-charge / over-discharge / balance
BAT	Pro Range INR 18650 pack	11.1 V (3S), 5000 mAh	1	Backup energy store
U1	MT3608 step-up booster	2 A max · set OUT = 12 V	1	Battery → stable 12 V on backup
U2	LM2596S-5.0 step-down buck	5 V fixed, 3 A	1	12 V rail → regulated 5 V
D1, D2	Schottky diode	SS54 / 1N5822 — 40 V, ≥3 A	2	Automatic OR-ing / switchover
F1	Fuse + holder	3 A	1	Input protection
SW1 · LED1-2 · R1-2	Toggle switch, 3 mm LEDs, 1 kΩ	—	opt.	On/off + mains & backup indicators

05 — CONNECTION GUIDE

Terminal-by-terminal wiring

Follow these in order. “+” wires are red, “–” (ground) wires are black.

#	FROM	TO	WIRE
1	Adapter +12 V	F1 → SW1 → 12 V node	red
2	12 V node	Charger IN+ and D1 anode	red (splits to two)
3	Charger OUT+ / OUT–	BMS P+ / P–	red / black
4	BMS B+ / B– (+ balance)	Battery pack + / – (and taps)	per BMS guide
5	BMS P+	MT3608 IN+ (IN– → GND)	red / black
6	MT3608 OUT+	D2 anode	red
7	D1 cathode + D2 cathode	join = 12 V RAIL	red
8	12 V RAIL	LM2596 IN+ (IN– → GND)	red / black
9	LM2596 OUT+ / OUT–	System +5 V / GND	red / black
10	All “–” terminals	Common GND bus	black

06 – SETUP & SAFETY

Before you power it on

Calibration steps

1. **Set the MT3608 first, before connecting the battery to the rail.**
Power it from the battery, turn its trim-pot until OUT reads **~11.8–12.0 V** on a multimeter — just below the mains rail so it only takes over when mains drops.
2. Confirm the charge module is set/fixed to **12.6 V** (3S). The LM2596S-5.0 is fixed 5 V — no adjustment.
3. Check every diode direction: the stripe (cathode) faces the **12 V rail** side on both D1 and D2.

NOTE · CHARGE LEVEL

A plain **12 V** adapter charges the pack to only ~12.0 V (≈90%). For a true 100% (12.6 V) charge, feed the charge module from a **12.6–14 V** adapter instead.

SAFETY · LI-ION

Never bypass the BMS. Never charge above 12.6 V. Fuse the input, avoid shorting the pack, and keep cells away from heat. Double-check polarity before first power-up.

4. Tie all module grounds together first, then add the positives.

Status indicators (optional)

● **LED1 — Mains** on whenever the adapter is supplying power. ● **LED2 — Backup** on when the battery is carrying the load.

Most charge modules already include a charging/full LED on board, so a separate charge indicator isn't needed.

Reference design for a battery-backed 5 V supply. Verify all polarities, voltages and module ratings against the actual modules you buy before applying power. Li-ion packs are hazardous if mishandled — this diagram is not a substitute for proper battery-safety practice.